Ofqual QAN	**600/0406/X**
Quartz ID	**1156, 1157**
Qualification	**Level 4 Diploma in Computing – Level 4 Diploma in Computing with Business Management**

Assessment	**Computer Network**
Assessment Type	**Examination**
Date	**Wednesday 5th March 2025**
Time	**9.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

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Answer ALL questions
Do NOT copy the questions to your answer booklet

Marks

Question 1

a) Explain in detail the purpose of the Data Link Layer in the OSI model **and** discuss its relationship to other layers. **5**

b) Name ONE (1) routed protocol **and** explain its function. **5**

Total 10 marks

Question 2

a) Explain in detail how a packet filtering firewall works. **4**

b) At which OSI layer does a packet filtering firewall work? **1**

c) Explain in detail the purpose of network security policies **and** outline TWO (2) examples. **5**

Total 10 marks

Question 3

List **and** explain FIVE (5) security threats to a network. **10**

Total 10 marks

Question 4

a) Explain the key characteristics of a client-server network **and** discuss its advantages over a peer-to-peer network. **4**

b) Explain in detail the difference between a physical network topology **and** a logical network topology. **6**

Total 10 marks

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Question 5

a) Explain in detail the function of a **switch** in a network. You should clearly identify the OSI layer at which it operates. **5**

b) Explain in detail the function of a **router** in a network. You should clearly identify the OSI layer at which it operates. **5**

Total 10 marks

Question 6

Star and mesh are TWO (2) common network topologies. Compare **and** contrast these topologies by completing the following:

a) Draw a diagram to illustrate each of them. **4**

b) Describe their relative advantages **and** disadvantages. **4**

c) Identify suitable applications for each topology. **2**

Total 10 marks

Question 7

Explain the process of installing **and** configuring a firewall on an internet-connected system. **10**

Your answer should include the SIX (6) key steps and considerations

Total 10 marks

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Question 8

a) Describe the FOUR (4) components of a VoIP system. **4**

b) Explain how these components work together. **2**

c) Discuss FOUR (4) security issues associated with VoIP. **4**

Total 10 marks

Question 9

Network Security is often defined in terms of THREE (3) key objectives expressed by the acronym CIA.

a) State **and** explain each of these THREE (3) objectives. **6**

b) Give examples of how technology can help achieve these objectives **4**

Total 10 marks

Question 10

a) Define what is meant by a VPN. **3**

b) Explain clearly **and** succinctly how a VPN works. **5**

c) Identify TWO (2) protocols used in a VPN **2**

Total 10 marks

End of paper

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